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MCY 650  
Reading Assignment  
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1. What are the 4 focuses and benefits of footprinting?

**Gathering network information:** Identifying the IP addresses, domain names, and network infrastructure used by the target. This helps understand the layout and structure of the target's network

**Collect system information:** Gathering details about the operating systems, software, and services running on the target systems. This allows the identification of potential vulnerabilities related to specific software versions or configurations.

**Understanding organizational structure:** Collecting data about the target organization, such as employee names, emails, and business operations. This enables social engineering attacks and helps in crafting phishing attempts.

**Assessing security posture:** Understanding the security mechanisms in place, such as firewalls, IDS/ IPS, and access controls. This provides insights into how well-protected the target is, allowing attackers to identify weak points or defenders to strengthen them.

2. What is the difference between passive footprinting and active footprinting?

**Passive footprinting:** Involves gathering information about a target without directly interacting with the target system or network. The goal is to remain undetected while collecting as much data as possible. e.g. social engineering, whois queries, DNS enumeration, public information

**Active footprinting:** Involves directly interacting with the target's systems, networks, or devices to gather information. This approach is more invasive and may be detected by the target. e.g. port scanning, ping sweeping, traceroute

3. Classify following activities into passive footprinting and active footprinting:

Dumpster diving **Passive**

Social engineering **Passive**

DNS probe **Active**

Phone calls to target company **Active**

Browsing company's website **Passive**

4. What is Zone Transfer?

Zone transfer is the replication of DNS records (maintained and managed by SOA for a namespace) to other DNS servers. Zone transfers are a necessary part of DNS operation but must be properly secured to prevent unauthorized access to critical DNS information.

5. Find the following DNS record for nku.edu domain using host command.

SOA, NS, MX, A

```
quezkemoni@Dquezs-MacBook-Pro ~ % nslookup
```

```
> set query=SOA
```

```
> nku.edu
```

```
Server: 10.110.10.12
```

```
Address: 10.110.10.12#53
```

```
Non-authoritative answer:
```

```
nku.edu
```

```
origin = ns5.nku.edu
```

```
mail addr = postmaster.nku.edu
```

```
serial = 2024082101
```

```
refresh = 28800
```

```
retry = 7200
```

```
expire = 604800
```

```
minimum = 86400
```

```
Authoritative answers can be found from:
```

```
> set query=MX
```

```
> nku.edu
```

```
Server: 10.110.10.12
```

```
Address: 10.110.10.12#53
```

```
Non-authoritative answer:
```

```
nku.edu mail exchanger = 0 nku-edu.mail.protection.outlook.com.
```

```
Authoritative answers can be found from:
```

```
nku-edu.mail.protection.outlook.com internet address = 52.101.9.17
```

```
nku-edu.mail.protection.outlook.com internet address = 52.101.11.7
```

```
nku-edu.mail.protection.outlook.com internet address = 52.101.11.15
```

```
nku-edu.mail.protection.outlook.com internet address = 52.101.42.13
```

```
> set type=NS
```

```
> nku.edu
```

```
Server: 10.110.10.12
```

```
Address: 10.110.10.12#53
```

```
Non-authoritative answer:
```

```
nku.edu nameserver = ns3.nku.edu.
```

```
nku.edu nameserver = ns4.nku.edu.
```

```
nku.edu nameserver = nsazdr.nku.edu.
```

```
Authoritative answers can be found from:
```

```
ns3.nku.edu internet address = 192.122.237.203
```

```
ns4.nku.edu internet address = 192.122.237.204
```

```
nsazdr.nku.edu internet address = 52.165.81.39
```

```
> set type=A
```

```
> nku.edu
```

```
Server: 10.110.10.12
```

```
Address: 10.110.10.12#53
```

```
Non-authoritative answer:
```

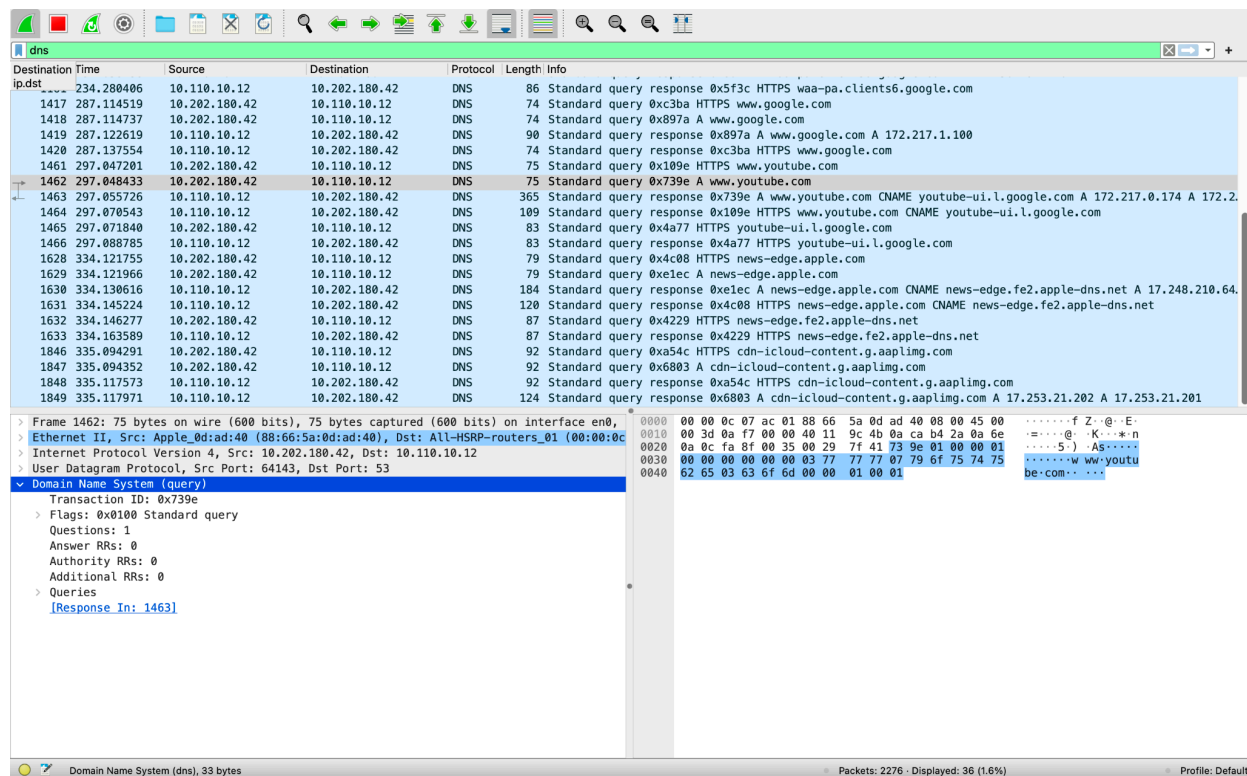
```
Name: nku.edu
```

```
Address: 20.15.228.104
```

6. Usually what transport layer protocol does DNS name lookup and Zone transfer use respectively? Use Wireshark to capture the any DNS name lookup data and show your screenshot here.

DNS name lookup usually uses UDP port 53 for DNS queries because it is faster and requires fewer resources.

Zone Transfer uses TCP port 53 for DNS zone transfer because it requires more reliability for data transmission of large sums of DNS data between servers.



| Destination | Time       | Source        | Destination   | Protocol | Length | Info                                                                                                   |
|-------------|------------|---------------|---------------|----------|--------|--------------------------------------------------------------------------------------------------------|
| ipdst...    | 234.280406 | 10.110.10.12  | 10.202.180.42 | DNS      | 86     | Standard query response 0x5f3c HTTPS waa-pa.clients6.google.com                                        |
| 1417        | 287.114519 | 10.202.180.42 | 10.110.10.12  | DNS      | 74     | Standard query 0xc3ba HTTPS www.google.com                                                             |
| 1418        | 287.114737 | 10.202.180.42 | 10.110.10.12  | DNS      | 74     | Standard query 0x897a A www.google.com                                                                 |
| 1419        | 287.122619 | 10.110.10.12  | 10.202.180.42 | DNS      | 90     | Standard query response 0x897a A www.google.com A 172.217.1.100                                        |
| 1420        | 287.137554 | 10.110.10.12  | 10.202.180.42 | DNS      | 74     | Standard query response 0xc3ba HTTPS www.google.com                                                    |
| 1461        | 297.047201 | 10.202.180.42 | 10.110.10.12  | DNS      | 75     | Standard query 0x109e HTTPS www.youtube.com                                                            |
| 1462        | 297.048433 | 10.202.180.42 | 10.110.10.12  | DNS      | 75     | Standard query 0x739e A www.youtube.com                                                                |
| 1463        | 297.055726 | 10.110.10.12  | 10.202.180.42 | DNS      | 365    | Standard query response 0x739e A www.youtube.com CNAME youtube-ui.l.google.com A 172.217.0.174 A 172.2 |
| 1464        | 297.070543 | 10.110.10.12  | 10.202.180.42 | DNS      | 109    | Standard query response 0x109e HTTPS www.youtube.com CNAME youtube-ui.l.google.com                     |
| 1465        | 297.071840 | 10.202.180.42 | 10.110.10.12  | DNS      | 83     | Standard query 0x4a77 HTTPS youtube-ui.l.google.com                                                    |
| 1466        | 297.088785 | 10.110.10.12  | 10.202.180.42 | DNS      | 83     | Standard query response 0x4a77 HTTPS youtube-ui.l.google.com                                           |
| 1628        | 334.121755 | 10.202.180.42 | 10.110.10.12  | DNS      | 79     | Standard query 0x4c08 HTTPS news-edge.apple.com                                                        |
| 1629        | 334.121966 | 10.202.180.42 | 10.110.10.12  | DNS      | 79     | Standard query 0xe1ec A news-edge.apple.com                                                            |
| 1630        | 334.130616 | 10.110.10.12  | 10.202.180.42 | DNS      | 184    | Standard query response 0xe1ec A news-edge.apple.com CNAME news-edge.fe2.apple-dns.net A 17.248.210.64 |
| 1631        | 334.145224 | 10.110.10.12  | 10.202.180.42 | DNS      | 120    | Standard query response 0x4c08 HTTPS news-edge.apple.com CNAME news-edge.fe2.apple-dns.net             |
| 1632        | 334.146277 | 10.202.180.42 | 10.110.10.12  | DNS      | 87     | Standard query 0x4229 HTTPS news-edge.fe2.apple-dns.net                                                |
| 1633        | 334.163589 | 10.110.10.12  | 10.202.180.42 | DNS      | 87     | Standard query response 0x4229 HTTPS news-edge.fe2.apple-dns.net                                       |
| 1846        | 335.094291 | 10.202.180.42 | 10.110.10.12  | DNS      | 92     | Standard query 0xa54c HTTPS cdn-icloud-content.g.aapling.com                                           |
| 1847        | 335.094352 | 10.202.180.42 | 10.110.10.12  | DNS      | 92     | Standard query 0x6803 A cdn-icloud-content.g.aapling.com                                               |
| 1848        | 335.117573 | 10.110.10.12  | 10.202.180.42 | DNS      | 92     | Standard query response 0xa54c HTTPS cdn-icloud-content.g.aapling.com                                  |
| 1849        | 335.117971 | 10.110.10.12  | 10.202.180.42 | DNS      | 124    | Standard query response 0x6803 A cdn-icloud-content.g.aapling.com A 17.253.21.202 A 17.253.21.201      |

> Frame 1462: 75 bytes on wire (600 bits), 75 bytes captured (600 bits) on interface en0, ...  
> Ethernet II, Src: Apple\_0d:ad:40 (88:66:5a:0d:ad:40), Dst: All-HSRP-routers\_01 (00:00:00:00:00:00) ...  
> Internet Protocol Version 4, Src: 10.202.180.42, Dst: 10.110.10.12 ...  
> User Datagram Protocol, Src Port: 64143, Dst Port: 53 ...  
v Domain Name System (query)  
Transaction ID: 0x739e  
Flags: 0x0100 Standard query  
Questions: 1  
Answer RRs: 0  
Authority RRs: 0  
Additional RRs: 0  
Queries  
[Response In: 1463]

7. whois is a command that can help you find info about a specific domain. Use whois to find out domain registry info on a domain of your choice.

Domain Name: **google.com**

Registry Domain ID: 2138514\_DOMAIN\_COM-VRSN

Registrar WHOIS Server: whois.markmonitor.com

Registrar URL: <http://www.markmonitor.com>

Updated Date: 2024-08-02T02:17:33+0000

Creation Date: 1997-09-15T07:00:00+0000

Registrar Registration Expiration Date: 2028-09-13T07:00:00+0000

Registrar: MarkMonitor, Inc.

Registrar IANA ID: 292

Registrar Abuse Contact Email: [abusecomplaints@markmonitor.com](mailto:abusecomplaints@markmonitor.com)

Registrar Abuse Contact Phone: +1.2086851750

8. If you know the IP address of a server of a company, say, a WWW server, you can find out the network range of the company at [www.arin.net](http://www.arin.net). If you enter the IP address, it will tell you the network range. Show the network range of a company of your choice. Note that the network shown on the site may show the range owned by a cloud service if the server is hosted on cloud. Google.com [8.8.8.8] Net Range: 8.8.8.0 - 8.8.8.255

What is footprinting? What do you do during the footprinting phase?

Active Footprinting vs Passive Footprinting

DNS footprinting

What is DNS poisoning?

What is the function of five regional Internet registries?

What are the DNS footprinting tools? What results can they give you?

What is Google hacking?

What are Web spiders used for?